

Parissa Alavi, PhD

Mechanical & Materials Engineer
Structural Mechanics, FEA & Metallic Materials

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French citizen (Swiss B permit) — Lausanne, Switzerland

Highlights

Structural & Computational Mechanics: 5+ years of experience in nonlinear mechanics, finite-element and multiscale modeling, plasticity, contact mechanics, and numerical simulation of metallic systems.

Metallic Materials Expertise: Strong metallurgy background spanning steels and aluminium alloys, including deformation, heat treatment, microstructure, mechanical testing, fatigue, wear, and failure analysis.

Industrial R&D with Novelis: Completed a 4-year EPFL–Novelis Switzerland research project developing and validating physics-based models for industrial aluminium sheet-metal processing.

Project Execution: [Agile Project Management Certified](#).

Technical & Engineering Skills

Structural Mechanics & FEA: Finite Element Method (FEM), nonlinear structural analysis, static analysis, modal analysis, dynamic analysis, continuum mechanics, plasticity, contact mechanics, thermomechanical behavior, material failure and multiscale modeling.

Metallic Materials & Manufacturing: Aluminium alloys, stainless steels, deformation and metal-forming processes, heat treatment, microstructure evolution, fatigue, wear, fracture and failure analysis.

Simulation & CAD Tools: Abaqus/CAE, COMSOL Multiphysics, SolidWorks, Autodesk Fusion 360, LAMMPS; experience adapting rapidly to new engineering simulation and CAD environments.

Numerical Methods & Engineering Computing: FEM, BEM, DEM, iterative solvers, conjugate-gradient methods, numerical optimization and physics-based model development.

Programming & Automation: Python (NumPy, SciPy), C++, MATLAB, Bash, Git and Linux; development and automation of engineering simulation workflows.

Experimental & Materials Characterization: Mechanical tensile testing, tribological testing, SEM, optical microscopy, XRD, heat-treatment processes and experimental model validation.

Industrial R&D & Professional Experience

Research Assistant (PhD Fellow) – Mechanical & Materials R&D

EPFL, in partnership with Novelis Switzerland

2021–2026

Industrial research on mechanical behavior, deformation and surface interactions in aluminium sheet-metal forming

Developed physics-based mechanical models for deformation, contact and surface evolution in industrial aluminium sheet-metal processing.

Built and validated multiscale numerical models combining continuum mechanics, plastic deformation, contact mechanics and surface characterization.

Applied FEM, BEM and DEM approaches to investigate mechanical interactions between metallic surfaces, lubricant pockets and third-body particles under industrial loading conditions.

Processed experimental surface measurements and industrial datasets to validate numerical predictions against physical material behavior.

Collaborated directly with Novelis R&D and manufacturing engineers to translate industrial metal-forming problems into predictive simulation tools and engineering recommendations.

Developed Python and C++ numerical tools and automated simulation workflows for large-scale mechanical analyses.

Research Assistant – Metallurgy & Mechanical Behavior

Sharif University of Technology

2018–2020

Project: Deformation, heat treatment and microstructure evolution of 304L stainless steel

Investigated deformation and thermomechanical processing of 304L stainless steel, including rolling, recrystallization, martensitic transformation and heat treatment.

Developed and validated a coupled Finite Element–cellular automata framework linking mechanical deformation, temperature and microstructural evolution.

Performed tensile testing, heat treatments and microstructural characterization to determine material response and validate numerical models.

Developed surrogate numerical models to accelerate prediction of metallurgical phase transformations.

Research Assistant – Metallic Component Failure Analysis

Amirkabir University of Technology – Railway Sector Collaboration

2016–2017

Project: Failure analysis and lifetime improvement of heavy railway components

Investigated fatigue and failure mechanisms in welded railway wheels using mechanical, metallurgical and microstructural analysis.

Assessed the effects of welding and thermomechanical processing on structural integrity, wear resistance and service life.

Developed optimized heat-treatment procedures to improve mechanical performance and component durability.

Industrial Internship

Fracture Analysis & Manufacturing Quality Intern

Robot Machine Factory – Renault supply chain

Jun–Sep 2014

Performed shop-floor failure investigations and quality-control assessments on manufactured automotive components.

Analyzed fatigue and fracture-related defects and evaluated component conformity with manufacturing specifications.

Worked across engineering, production-floor and quality-control teams, gaining direct exposure to manufacturing constraints and inspection workflows.

Education

PhD in Mechanical Engineering

2021–2026

École Polytechnique Fédérale de Lausanne (EPFL) (*Industrial research collaboration with Novelis Switzerland*)

Achievement: Thesis ranked in top 8% and nominated for the EPFL Best Thesis Award.

M.Sc. in Materials Science and Engineering

2018–2020

Sharif University of Technology

Achievement: Ranked 1st of 12 graduates.

B.Sc. in Materials Science and Engineering

2012–2017

Amirkabir University of Technology

Achievement: Ranked 3rd out of 60 graduates.

Technical Leadership & Knowledge Transfer

Supervision: Mentored and supervised 4+ Master's-level engineering semester projects, tracking engineering milestones, simulation quality, and documentation.

Teaching & Knowledge Transfer: Served as Teaching Assistant for *Numerical Modeling of Solids and Structures (FEM)* and *Scientific Programming for Engineers*, delivering hands-on instruction in software development, algorithmic logic, and continuum mechanics.

Event Execution: Organized the EPFL Open Science Summer School ([Event Website](#)), coordinating logistics and managing project workflows for international technical speakers and attendees.

Languages

English: Full Professional Proficiency (C1/C2)

French: Professional Working Proficiency (B2)

German: Elementary (A2)

Persian: Full Professional Proficiency (C2)

Selected Publications

P. Alavi, L. Rocchi, C. Leppin, J.-F. Molinari, G. Anciaux (2026). *A multiscale model of friction considering the influence of third-body wear particles*. Wear.

P. Alavi, G. Anciaux, L. Rocchi, J. Richard, J.-F. Molinari (2026). *Numerical modeling of rough contact interfaces with trapped compressive liquid pockets*. Tribology International.

P. Alavi, S. Serajzadeh (2020). *Simulation of recrystallization and martensite reversion in 304L austenitic stainless steel after multi-pass rolling*. Multiscale and Multidisciplinary Modeling, Experiments and Design.